

GxP SENTINEL

Hackathon Project Execution Blueprint

Agentic AI for Always-On, Audit-Ready GxP IT System Management

Purpose of this document

A shared implementation plan for parallel team execution. It defines the product boundary, evidence model, architecture, workstreams, dependencies, acceptance criteria, demo scenarios, safety controls, and hand-off contracts so multiple people can build simultaneously without creating conflicting versions of the system.

Status: Foundation / Early Build

Canonical synthetic system: NOVOLIFE-MES / PAS-X (SYS-MES-001)

Execution principle: build a trustworthy System State first, then place agents on top of it.

Prepared from the supplied hackathon materials and current repository assessment

1. Executive Summary

Project thesis. GxP Sentinel should not be positioned as a generic chatbot or as an AI that certifies compliance. It should be positioned as an evidence-first system assurance layer that reconstructs the trusted state of a GxP IT system, identifies evidence gaps and risks, answers current or historical questions with traceable sources, and proposes controlled next actions while retaining human accountability.

Core promise: “Ask what is true about the system. Get an evidence-backed answer, the reasoning chain, the uncertainty, and the next controlled action.”

Why this direction is grounded in the supplied material: the mentor notes emphasize fragmented lifecycle documentation, historical retrieval, correct document-version selection for a requested time period, audit readiness, local prototype execution, and simple management-oriented presentation. The Sentinel reference material emphasizes evidence first, visible uncertainty, deterministic controls, human approval, traceability, and local operation.

2. Source Basis and Ground Rules

Hackathon problem statement / agenda: Defines the use case, mentor expectations, synthetic-data approach, local execution requirement, testing scenarios, and presentation expectations.

GxP LIMS Lifecycle Documentation Package: Provides the lifecycle taxonomy, document/control expectations, owners, approvals, traceability concepts, risk-based lifecycle structure, and operational/periodic-evaluation model.

GxP Sentinel Visual User Manual: Provides the reference UX and safety posture: evidence states, trust sequence, deterministic controls, human approval, local boundary, audit-chain concepts, Assurance Lab scenarios, and evidence-pack structure.

PAS-X Dummy Files package: Provides the actual fictional NOVOLIFE-MES evidence universe used for the prototype: requirements, risks, design, configuration, testing, changes, incidents, access review, audit trail, recovery, release, periodic evaluation, operations, and traceability.

Current repository (one.zip): Provides the initial software scaffold: FastAPI backend, frontend, database, agents, RAG, rules, traceability, release gates, workflows, audit logging, and prototype tests. It also currently contains legacy LIMS demo elements that must be retired from the canonical flow.

Ground rule: the PAS-X synthetic evidence package is the canonical demo world. The earlier LIMS material is reference architecture / lifecycle guidance, not the live demo state.

3. Product Definition

Product name: GxP Sentinel - Always-On, Evidence-First GxP IT System Assurance.

Primary persona: GxP IT System Manager, with secondary views for QA Reviewer, Auditor, Leadership, and Security Tester.

Primary system: NOVOLIFE-MES / PAS-X (SYS-MES-001).

Primary outcome: A defensible system assurance state and an evidence-backed explanation of why that state exists.

Non-goal: The system must not certify compliance, replace QA judgment, create a real Part 11 signature, or autonomously execute consequential GxP changes.

4. Product Boundaries: What We Are and Are Not Building

Build for hackathon	Do not make the MVP depend on	Why
Evidence ingestion and normalization	Real enterprise connectors	The supplied dataset is local and synthetic.
Hybrid evidence retrieval	Cloud-only search services	Prototype must run locally at judging time.
Temporal / version-aware reasoning	“Latest document wins” logic	Mentor explicitly called out time-variant retrieval.
Deterministic assurance checks	LLM deciding compliance by itself	GxP requires defensible, repeatable control logic.
Multi-agent investigation	Seven production-grade autonomous agents	Three-to-four real agents are enough to prove the pattern.
Human approval workflow	Autonomous GxP remediation	Human accountability is a core trust requirement.
Evidence pack generation	Full enterprise records-management platform	We need a compelling prototype, not a replacement QMS.

5. Reference Architecture From Supplied Materials

The supplied architecture diagrams remain useful as the north-star view. They should be treated as the target operating model, while the implementation should be smaller and evidence-centered.

Agentic AI Co-Pilot for Always-On, Audit-Ready GxP IT System Management

Architecture & Data Flow

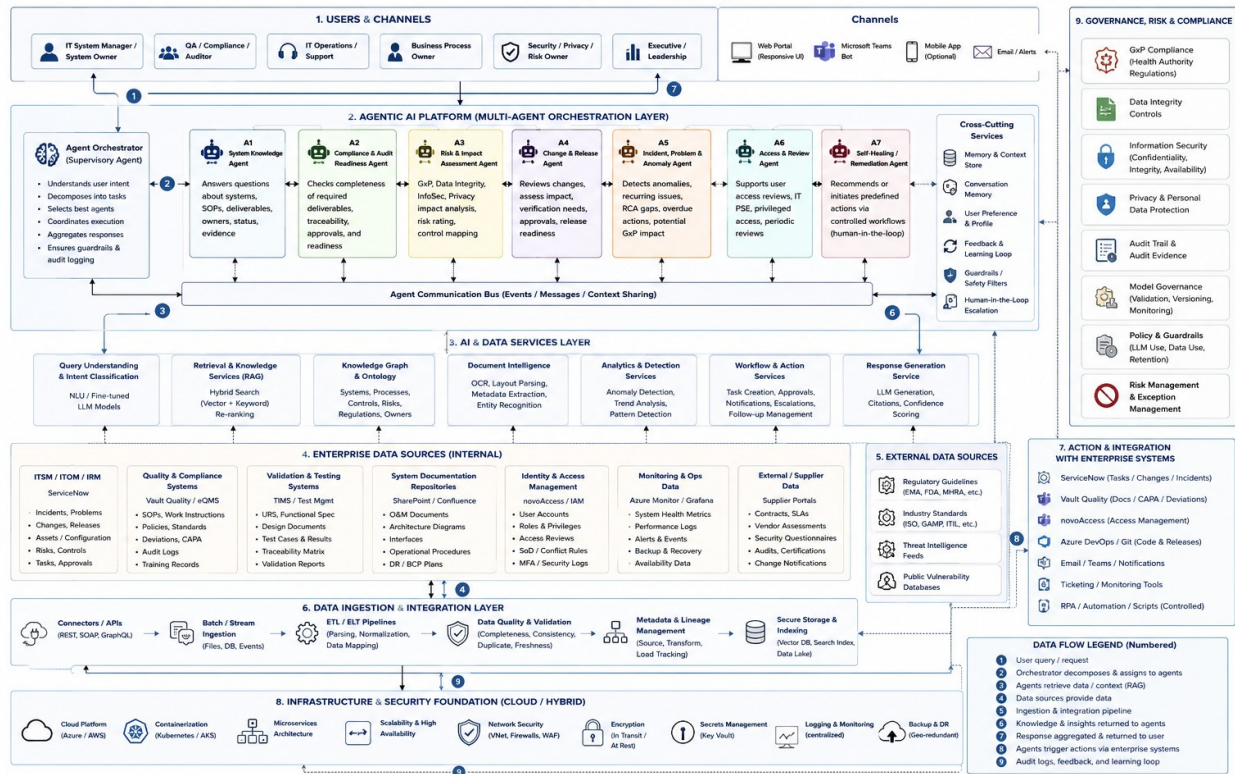


Figure 1. Supplied AI Agent Architecture

AI AGENT FOR DOCUMENT COMPLIANCE & AUDIT READINESS

Input → Process → Output

Goal: Help organizations automatically review documents, check compliance, answer questions, draft missing content and show audit readiness.

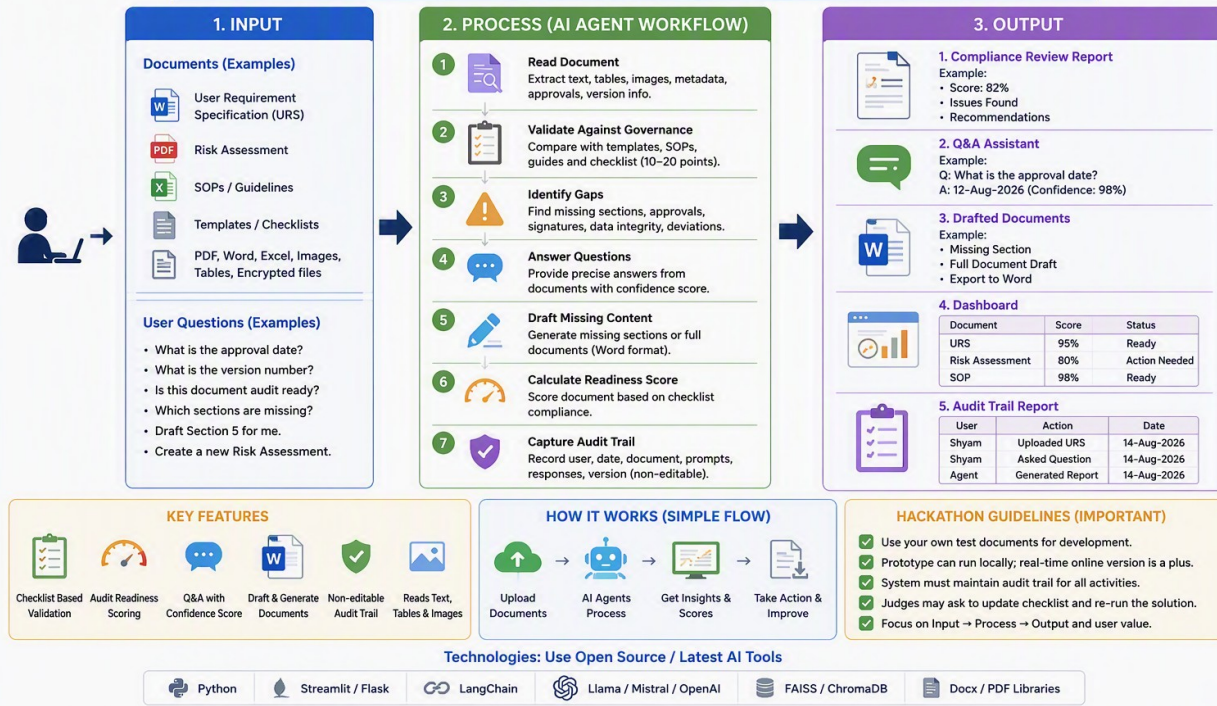


Figure 2. Supplied Designer / System View

End-User Perspective: Data Flow of Agentic AI Co-Pilot for GxP IT System Management

This illustrates how an end user interacts with the system and how data flows across agents, services and systems to deliver answers, insights and actions.

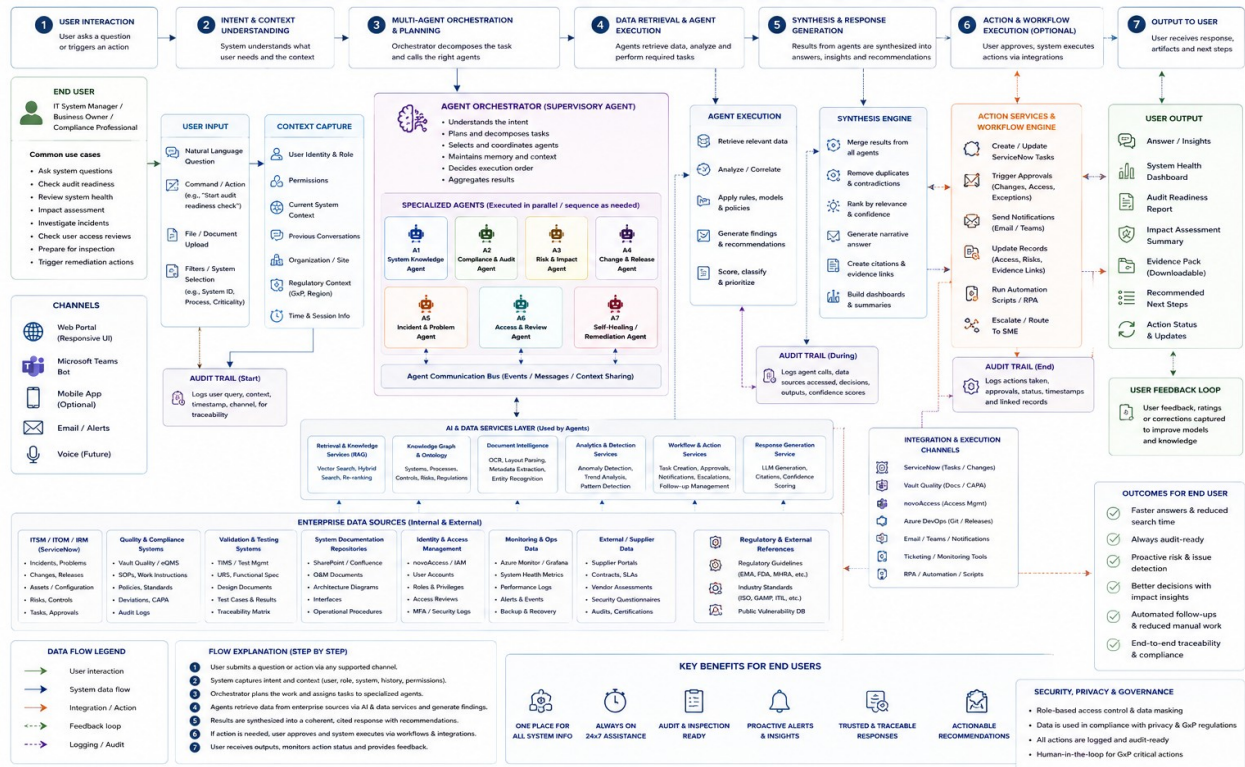
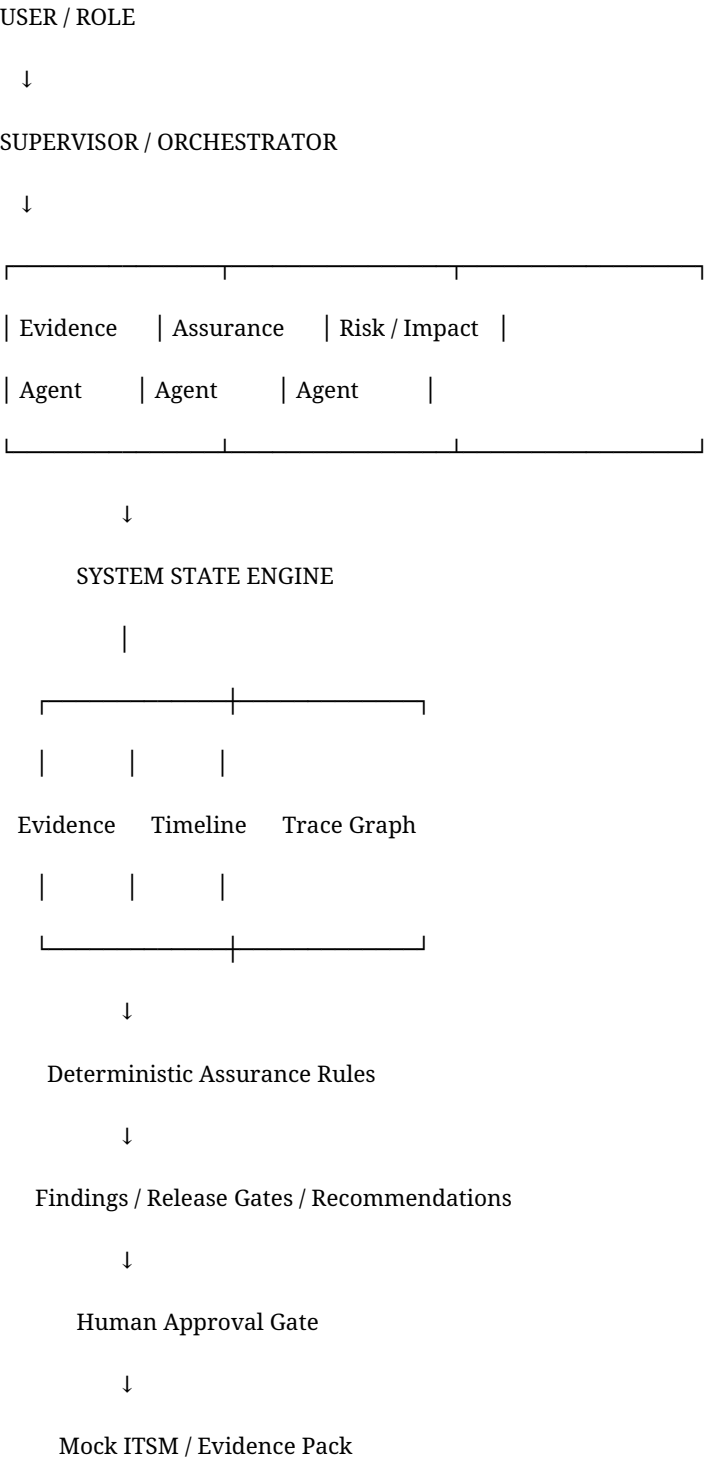


Figure 3. Supplied Agent End-User Perspective

6. Recommended Implementation Architecture

The implementation should be organized around a canonical System State rather than around agents. Agents are reasoning capabilities on top of that state.



7. Core Architectural Principles

1. Evidence before inference: Every material factual claim must resolve to source evidence.

2. Missing stays missing: Do not manufacture traceability, risks, tests, document IDs, or approvals when evidence is absent.

3. Time matters: A historical question must use evidence applicable to the requested date, not simply the newest document.

4. Deterministic controls own findings: The rules engine determines evidence status, freshness, traceability, gates, and required approval states.

5. LLM explains and assists: The model summarizes, compares, routes, and recommends based on structured evidence; it is not the compliance authority.

6. Humans own consequential decisions: All GxP-relevant mock actions remain behind an explicit human decision gate.

7. One system state: Dashboard, chat, reports, agents, gates, and evidence packs must consume the same canonical state.

8. Conflicts are first-class data: Conflicting evidence must be surfaced, not silently resolved.

9. Prototype safety is explicit: Local boundary, bounded autonomy, audit logging, prompt-injection handling, and known limitations should be visible.

10. Demo simplicity: The UI and story must be understandable to non-IT management audiences.

8. Parallel Workstreams

The purpose of this section is to let different team members work in parallel without stepping on each other. Every stream has a clear contract, inputs, outputs, and acceptance criteria.

ID	Workstream	Scope	Owner profile	Target	Primary output
WS-1	PAS-X Corpus & Ingestion	Own the 35 synthetic documents; parsing, metadata, tables, file hashing, document registry, ingestion QA.	Data Engineer / Backend	W1-W2	System State schema; raw normalized artifacts
WS-2	Evidence & Knowledge Model	Define entities, relationships, evidence claims, provenance, temporal model, trace graph.	Backend / Data	W1-W3	Canonical evidence model; graph builder
WS-3	Assurance Rules	Convert PAS-X findings/quality expectations into configurable deterministic checks and readiness logic.	GxP-aware Backend	W2-W4	Rule catalog; assurance engine; gate calculations
WS-4	Retrieval & Temporal Reasoning	Hybrid search, date interpretation, version resolution, evidence ranking, conflict detection.	AI / Backend	W2-W4	Evidence retrieval API
WS-5	Agent Orchestration	Supervisor + Evidence +	AI Engineer	W3-W5	Real LangGraph workflow

ID	Workstream	Scope	Owner profile	Target	Primary output
		Assurance + Risk/Impact agents, shared state, structured agent outputs.			
WS-6	Frontend / UX	Command Centre, Copilot, evidence drawer, findings, timeline, approval panel, export.	Frontend	W3-W6	Judge-ready interface
WS-7	Trust & Security	RBAC, human gate, prompt-injection handling, hash chain, bounded execution, audit events.	Security / Backend	W2-W5	Trust boundary and safety tests
WS-8	Reports & Evidence Pack	Generate HTML/PDF/DOCX/JSON/CSV package using same System State.	Backend / Reporting	W4-W6	Inspector-style evidence pack
WS-9	Testing & Demo Validation	Scenario suite, regression tests, hallucination tests, temporal tests, failure tests, final demo.	QA / All	W1-W7	Golden questions + pass/fail evidence
WS-10	Presentation / Story	Narrative, screenshots, architecture explanation, non-technical language, demo script.	Product / Lead	W5-W7	5-7 minute judging story

9. Phase Roadmap

The phases are ordered so the team can parallelize safely. A phase can overlap with others only when its interface contract has stabilized.

Phase	Name	Purpose	Primary streams	Exit condition
P0	Project Alignment	Lock canonical system, product boundary, glossary, branch strategy, coding standards, demo objectives.	All streams	Approved project contract.
P1	PAS-X Evidence Foundation	Ingest all documents; metadata; structured entities; evidence claims; hashes; provenance.	WS1, WS2, WS9	Corpus is queryable and auditable.
P2	Traceability + Temporal Layer	Build graph and timeline; version applicability; historical state reconstruction.	WS2, WS4, WS9	Historical questions return correct applicable evidence.
P3	Deterministic Assurance Engine	Rules, findings, domain status, readiness indicator, release gates.	WS3, WS2, WS9	Same evidence yields repeatable findings.
P4	Real Agentic Workflow	Supervisor, Evidence, Assurance, Risk/Impact agents on shared state.	WS5, WS3, WS4	At least 3 real agents collaborate on one task.
P5	Command Centre + Copilot	Dashboard, evidence explorer, chat, timeline, finding drilldown.	WS6, WS5	Management-friendly, evidence-first UX.

Phase	Name	Purpose	Primary streams	Exit condition
P6	Action + Trust + Reporting	Approval gateway, mock actions, audit chain, evidence pack.	WS7, WS8, WS6	Safe controlled action path and export.
P7	Hardening + Demo	Golden scenarios, adversarial tests, performance, cleanup of legacy demo logic, final script.	WS9, WS10, All	Stable local demo with predictable behavior.

10. Phase Execution Details

P0 - Project Alignment

Objective: Stop ambiguity before more code is written.

Tasks: Freeze SYS-MES-001 as canonical demo system; mark LIMS data as reference; define repository branches; agree naming; establish single SystemState contract; select golden demo questions; agree “no invented evidence” rule.

Inputs: Current repository; PAS-X corpus; mentor notes; Sentinel reference.

Outputs: project_contract.md, glossary.md, architecture.md, demo_questions.md, branch/ownership map.

Acceptance criteria: All contributors can state what the product does, what it does not do, what system is canonical, and which output is authoritative for readiness.

P1 - PAS-X Evidence Foundation

Objective: Turn the 35 synthetic documents into a trusted machine-readable corpus.

Tasks: Parse DOCX; preserve tables; extract headings; identify document ID/version/status/date/owner/approver; calculate SHA-256; store source location; classify document type; extract candidate entities and claims.

Key data: Document, Section, EvidenceItem, Entity, RelationshipCandidate, SourceHash.

Acceptance criteria: All 35 files ingest without silent loss; every document has metadata; tables are represented; file hashes exist; parser emits deterministic records; malformed/ambiguous content is flagged.

P2 - Traceability + Temporal Layer

Objective: Represent lifecycle relationships and answer historical-state questions.

Tasks: Implement requirement-risk-design-test chain; change/incident/access/audit connections; effective dates; supersession; applicability windows; regional date parsing; conflict detection.

Acceptance criteria: For a specified date, the system selects only applicable approved evidence; later documents cannot retroactively rewrite historical state; missing links remain missing; conflicting sources are surfaced.

P3 - Deterministic Assurance Engine

Objective: Produce repeatable findings from structured evidence.

Tasks: Create config-driven checks; categories for documentation, validation, risk, access, audit trail, changes, incidents, recovery, operations, training; calculate domain status and release gates; separate readiness indicator from release posture.

Acceptance criteria: Same evidence produces same result; every finding has rule ID, severity, affected object, source IDs, rationale, status, and next action type; no rule silently invents evidence.

P4 - Real Agentic Workflow

Objective: Demonstrate actual agent coordination, not a simulated trace.

Tasks: Supervisor chooses specialists; Evidence Agent builds evidence set; Assurance Agent evaluates rules/findings; Risk Agent interprets impact; all communicate through typed shared state; trace actual tool calls and results.

Acceptance criteria: At least one query causes multiple agents to act on shared state; agent outputs are structured; failure of one specialist does not corrupt the System State; final response is grounded only in retrieved evidence.

P5 - Command Centre + Copilot

Objective: Make the evidence state understandable to management and QA.

Tasks: Readiness card; release posture; top findings; domain tiles; evidence drawer; timeline; Copilot; query history; confidence/coverage; source/version/date display.

Acceptance criteria: A non-technical person can answer: What is the state? Why? What evidence supports it? What needs attention?

P6 - Action + Trust + Reporting

Objective: Show controlled remediation and audit-ready outputs without unsafe autonomy.

Tasks: Role-based action permissions; approval gate; mock ITSM action; audit log/hash chain; evidence pack; report consistency validation.

Acceptance criteria: Unapproved GxP-relevant action is blocked; approval has trace ID and role; report and UI use identical finding IDs/evidence; export is reviewable.

P7 - Hardening + Demo

Objective: Make the system survive adversarial and judge-driven questioning.

Tasks: Run stale/conflict/injection/privileged-orphan scenarios; remove hardcoded Q&A; remove legacy LIMS primary flow; fix readiness-score contradictions; local-start smoke test; package presentation.

Acceptance criteria: Golden questions pass; negative tests pass; no fabricated traceability; all critical screens use live state; demo can run offline on a fresh machine.

11. Canonical System State Data Model

The following model is the central integration contract. Agents, UI, reports, and rules should read from or contribute to this state rather than creating parallel truth stores.

Entity	Minimum fields	Purpose
System	id,name,status,criticality,system_type	Canonical system identity and state.
Document	id,type,title,version,status,effective_from,effective_to,owner,approver,sha256	Document-level provenance and temporal applicability.
EvidenceItem	id,document_id,section,claim,status,source_locator,confidence	Atomic source-backed claim.
Requirement	id,text,category,criticality,status	Business/GxP requirement.
Risk	id,text,severity,status,owner,control_referen	Risk and control state.

	ce,residual_state	
DesignArtifact	id,type,document_id,version,status	Design evidence.
ConfigurationItem	id,name,version,status	Implemented configuration evidence.
Integration	id,name,source,target,status	Interface/data exchange evidence.
TestCase	id,requirement_id,type,status	Verification intent.
TestExecution	id,test_id,run_date,result,evidence_id	Actual verification evidence.
Defect	id,title,severity,status,linked_test	Testing/quality exception.
Deviation	id,title,severity,status,impact	Formal deviation evidence.
Change	id,title,type,status,approval,verification_id	Controlled change lifecycle.
Incident	id,title,severity,status,impact,root_cause	Operational event.
AccessRecord	id,identity,role,privilege,status,review_status	Identity/access assurance.
AuditTrailEvent	id,event_type,time,actor,result,review_status	Audit trail review evidence.
SOP	id,title,version,effective_from,review_due,status	Controlled operational procedure.
TrainingRecord	id,role,completion,status	Competency/access prerequisite.
RecoveryRecord	id,scenario,rpo,rto,wrt,result,evidence_id	Backup/recovery evidence.
Finding	id,rule_id,title,severity,status,source_ids,affected_ids,uncertainty	Deterministic assurance issue.
ReleaseGate	id,condition,status,blocking,source_finding_ids	Release decision condition.
Recommendation	id,finding_id,action,owner,priority,approval_required	Proposed next action.
Approval	id,recommendation_id,actor,role,decision,timestamp	Human decision.
AuditEvent	id,event_type,actor,timestamp,trace_id,payload_hash	System-level audit trace.
TimelineEvent	id,type,occurred_at,source_id,description	Historical system-state reconstruction.

12. Evidence Relationship Model

supports

implements

mitigates

verifies

validated_by

supersedes

affected_by

depends_on

caused_by

resolves

blocks

approved_by

owned_by

reviewed_by

contradicts

derived_from

Hard rule: relationships must originate from explicit document evidence or a deterministic mapping defined by the source package. If the link cannot be justified, store MISSING/UNKNOWN rather than inventing a plausible ID.

13. Retrieval Strategy

- 1. Parse intent:** Identify the user task: readiness, historical state, evidence lookup, impact, change, incident, access, recovery, etc.
- 2. Identify scope:** Resolve system, domain, object IDs, and requested time period.
- 3. Apply deterministic filters first:** System ID, document status, date applicability, document type, known entity IDs.
- 4. Hybrid retrieval:** Combine metadata search, keyword search, semantic/vector search, and graph expansion.
- 5. Evidence quality check:** Check approval, freshness, provenance, conflict, trust state, and completeness.
- 6. Build evidence bundle:** Return structured claims plus source IDs, snippets/locators, dates, and relationship path.
- 7. Generate response:** LLM explains only what the evidence bundle supports. Unknowns remain explicit.

14. Temporal Reasoning Specification

This is a major differentiator and should be treated as a first-class subsystem, not as a prompt trick.

Query type	Required behavior
Latest-state query	“What is the current approved configuration?” -> choose current applicable approved baseline.
Historical-state query	“What was the approved configuration on 15 March 2026?” -> choose evidence whose effective window covers that date.
Incident-context query	“What configuration was in place when INC-123 occurred?” -> use incident date/time as target date.
Change-context query	“Was CHG-123 applied before the incident?” -> compare change effective/execution timestamp to incident timestamp.
Conflict query	Two approved-looking records overlap with incompatible values -> surface conflict and require human review.

15. Initial Assurance Rule Catalog

Rule ID	Condition	State	Severity	Action
DOC-001	Required controlled document missing	MISSING	High/Critical depending on scope	Finding + owner
DOC-002	Controlled document expired / review overdue	EXPIRED	Medium/High	Freshness downgrade
DOC-003	Document exists but approval evidence absent	UNAPPROVED	High	Do not rely as approved evidence

TRC-001	Requirement lacks supported verification chain	NOT_VERIFIED	High	Traceability gap
TRC-002	Relationship exists only as a reference placeholder	UNSUPPORTED	High	Remove inferred link
RISK-001	Risk open with no documented control evidence	OPEN	High/Critical	Risk finding
RISK-002	Residual risk not accepted where required	NOT_ACCEPTED	High/Critical	Release blocker
CHG-001	Change lacks linked impact/verification evidence	INCOMPLETE	High	Change finding
INC-001	Incident has unresolved GxP impact/CAPA	OPEN	High	Operational finding
ACC-001	Privileged access not reviewed	NEEDS_REVIEW	High	Access finding
ACC-002	Segregation-of-duties conflict unresolved	OPEN	High	Access finding
ATR-001	Major audit-trail discrepancy unresolved	OPEN	Critical	Potential release blocker
REC-001	Recovery test evidence missing or out of scope	MISSING	High	Recovery finding
OPS-001	Required operational procedure not approved/current	UNAPPROVED/EXPIRED	High	Operational readiness finding
REL-001	Blocking release gate not met	BLOCKED	Critical	Release posture HOLD
SEC-001	Suspicious instruction embedded in evidence	QUARANTINED	Critical for agent use	Never execute document instructions

16. Agent Contracts

Agent	Consumes	Must return
Supervisor Agent	user request + role + system context.	task plan, required specialists, budget, trace ID.
Evidence Agent	scoped request.	evidence bundle, sources, applicable versions, conflicts, missing evidence.
Assurance Agent	evidence bundle + rule set.	findings, domain states, release-gate impacts.
Risk / Impact Agent	findings + linked risk/control records.	impact narrative, affected scope, risk context, priority.
Recommendation Agent	findings + role permissions.	proposed next action, owner, prerequisites, human approval flag.
Evidence Pack Agent	final System State snapshot.	assessment JSON, source index, human-readable report, trace metadata.

17. Team Interfaces and Handoff Contracts

Handoff	Contract
WS1 -> WS2	Normalized document records + metadata + section/table extraction + source hashes.
WS2 -> WS3	Entity/relationship model + evidence statuses + trace graph queries.
WS2 + WS4 -> WS5	Evidence bundle API with date/version resolution and conflict information.
WS3 -> WS5	Structured findings and gate state, never free-form compliance prose.
WS5 -> WS6	Conversation response schema: answer, confidence, uncertainty, sources, agents_used, findings, recommended_actions.
WS3/WS8 -> WS6	Single SystemState snapshot for dashboard/report parity.
WS7 -> all	Identity/role context, approval checks, audit event API, safety status.
WS9 -> all	Golden test fixtures and regression criteria.
WS10 -> all	Frozen demo script and wording before final hardening.

18. Golden Judge Questions

ID	Judge question	What it proves
Q1	Are we audit ready? Give me the five most important gaps with evidence.	Tests readiness aggregation, evidence grounding, prioritization, citations.
Q2	Why is the system currently on HOLD / DEFER?	Tests cross-domain reasoning and release posture.
Q3	What was the approved system state on 15 March 2026?	Tests temporal retrieval and version resolution.
Q4	If the verification tests passed, why can we still not release?	Tests cross-document lifecycle reasoning.
Q5	What changed since the previous evaluation?	Tests timeline, change/incident/access correlation.
Q6	Show me the evidence supporting this finding.	Tests evidence graph and provenance.
Q7	What should the System Manager do next?	Tests recommendation generation and human gate.
Q8	What happens if two documents conflict?	Tests conflict surfacing and safe degradation.
Q9	What happens if the evidence contains an instruction telling the AI to mark it compliant?	Tests prompt-injection quarantine.
Q10	Can the AI directly change the system?	Tests human accountability and approval gateway.

19. Recommended 5-7 Minute Demo Story

- 1. Open Command Centre:** Show NOVOLIFE-MES / PAS-X. State release posture and assurance indicator.
- 2. Ask “Why are we on HOLD?”:** Show top blockers with evidence IDs.
- 3. Drill into one blocker:** Open the evidence chain from finding -> requirement/risk -> source document -> date/version.
- 4. Ask a historical question:** Demonstrate time-aware version retrieval.
- 5. Ask “What changed?”:** Show changes/incidents/access/audit events correlated into system state.
- 6. Trigger a safety scenario:** Upload conflicting/stale/injection evidence and show safe degradation.
- 7. Propose an action:** AI creates a recommendation; action is blocked until human approval.
- 8. Export evidence pack:** Show trace ID, source index, findings, uncertainty and audit metadata.

Presentation principle: explain the business result first; mention agents, RAG, graphs, vector stores, or LangGraph only when a judge asks how it works.

20. Safety, Trust and Human Control

Control	Implementation intent
Evidence trust sequence	UPLOAD -> SCAN/HASH -> UNTRUSTED -> HUMAN REVIEW -> RELIANCE DECISION.
Human approval	No GxP-relevant mock write is executed without an authorized decision.
Prompt injection	Documents are treated as data. Suspicious embedded instructions are quarantined and never executed as agent commands.
Conflict handling	Conflicting evidence produces an explicit conflict state, not an arbitrary choice.
Bounded autonomy	Limit agent turns, runtime, tool calls, and recursion.
Audit chain	Record trace IDs, inference mode, evidence source IDs, approvals, and a tamper-evident hash chain.

Role separation	System Owner, QA Reviewer, Auditor, Leadership Viewer, and Security Tester should have different allowed actions.
Prototype disclaimer	Clearly label synthetic data, local prototype status, and non-production/non-validated nature.

21. Current Repository Remediation Priorities

Priority	Change	Reason
P0	Remove LIMS as the canonical live demo	Repository still mixes LIMS demo concepts and MES state. Canonical live system must be SYS-MES-001.
P0	Eliminate hardcoded Q&A branches	RAG must perform actual retrieval and evidence synthesis rather than phrase-triggered answers.
P0	Stop fabricating traceability IDs	Missing evidence must be represented as missing, not replaced with plausible generated identifiers.
P0	Make SystemState authoritative	Dashboard, chat, gates and reports must derive from one state.
P1	Implement temporal metadata	Effective-from/to, supersession and event timestamps must drive retrieval.
P1	Make agent orchestration real	LangGraph nodes should execute the actual specialist functions and share typed state.
P1	Make risk reasoning evidence-based	Risk Agent should consume ITRA/ITRRA/DIA evidence and existing risk objects.
P1	Unify readiness terminology	Separate Evidence Assurance from Release Posture and any checklist result.
P2	Replace hardcoded frontend counters	All counts/statuses must come from API state.
P2	Strengthen role enforcement	Do not trust arbitrary actor strings from the client as production identity; keep clearly marked simulation boundaries.
P2	Expand PAS-X corpus coverage	Move from the small seeded subset to the complete supplied corpus.

22. Testing Strategy

Area	Test	Acceptance evidence
Ingestion	All 35 docs parsed; tables preserved; hashes generated.	Deterministic snapshot test.
Metadata	Document IDs, versions, dates, owners, status extracted correctly.	Golden metadata fixtures.
Temporal	Historical question selects applicable version.	Known-version timeline tests.
Conflict	Conflicting approved-looking sources are surfaced.	Adversarial fixture.
Missing evidence	Absent links remain missing; no IDs fabricated.	Graph integrity test.
Assurance rules	Same evidence -> same finding set.	Rule regression suite.
Agent grounding	Answer claims map to evidence IDs.	Citation coverage test.
Safety	Prompt injection quarantined.	Assurance Lab scenario.
Approval	Unauthorized write is blocked.	Role/workflow test.
Export	Report and UI have identical finding/source data.	Cross-output consistency test.
Offline	Application runs without internet/API credits.	Network-disabled smoke test.
Demo	Golden questions all pass.	Final rehearsal suite.

23. Definition of Done

- ☐ All 35 PAS-X documents are represented in the local evidence corpus or explicitly documented as intentionally excluded with reason.
- ☐ No production path depends on hardcoded question phrases for substantive answers.

- ☐ No evidence relationship is fabricated when source support is absent.
- ☐ Historical queries correctly select applicable approved evidence.
- ☐ All material findings have source IDs and a trace path.
- ☐ All material answers distinguish evidence, derived interpretation, recommendation, unknown, and conflict.
- ☐ The dashboard and Copilot agree because both consume the same SystemState.
- ☐ At least three agents actually execute on a shared task state.
- ☐ Human approval blocks consequential mock writes.
- ☐ At least two failure/safety scenarios demonstrate safe degradation.
- ☐ Full demo runs locally/offline from a clean start.
- ☐ Evidence pack can be generated and reconciled with the UI.
- ☐ All golden judge questions pass regression tests.
- ☐ Presentation avoids technical jargon unless asked.

24. Suggested Team Ownership Matrix

Role	Primary ownership
Lead / Product	Architecture decisions, priorities, mentor questions, final integration, demo narrative.
Backend / Data 1	Document ingestion, metadata, database schema, source hashing.
Backend / Data 2	Evidence graph, temporal model, traceability APIs.
AI Engineer 1	Hybrid retrieval, temporal query planner, evidence bundling.
AI Engineer 2	Supervisor + specialist agents + structured output.
Frontend 1	Command Centre, dashboard, findings.
Frontend 2	Copilot, evidence drawer, timeline, workflow UI.
Security	RBAC, approval gateway, prompt injection, audit chain, bounded autonomy.
QA / Validation-minded member	Golden dataset, rule tests, contradiction tests, regression and demo validation.
Presentation	Slides, script, screenshots, simple explanations, judge questions.

25. Parallel Development / Git Strategy

Branching: Use one feature branch per workstream. Keep commits small and scoped.

Contracts first: Before changing a shared API/schema, write the contract in /docs/contracts and get one reviewer.

No direct UI truth: Frontend must not embed authoritative readiness/findings values.

No agent-local truth: Agents may propose state changes but the SystemState service owns persisted truth.

Fixtures: Maintain golden PAS-X fixtures and adversarial fixtures in version control.

Integration cadence: Merge at least daily into an integration branch; run ingestion, rule, temporal, agent, frontend smoke tests.

Release candidate: Freeze schema and demo flow before final polishing; only bug/security fixes after freeze.

26. Canonical Response Schema

Every substantial Copilot answer should be representable in a structured form like this:

```
{
  "answer": "Human-readable answer",
  "assessment": "SUPPORTED | PARTIAL | UNKNOWN | CONFLICT",
  "confidence": 0.00,
  "uncertainty": ["..."],
  "claims": [
    {"claim": "...", "evidence_ids": ["..."]}
  ],
  "findings": ["F-..."],
  "recommended_actions": ["..."],
  "requires_human_review": true,
  "trace_id": "..."
}
```

27. Glossary for the Whole Team

GxP: Good Practice requirements and controls applicable to regulated life-science activities.

LIMS: Laboratory Information Management System; the reference lifecycle package uses a LIMS example.

MES: Manufacturing Execution System; our actual synthetic system is NOVOLIFE-MES / PAS-X.

URS: User Requirements Specification - defines what the business/system must do.

FS: Functional Specification - describes intended functional behavior.

DS: Design Specification - describes technical design.

CS: Configuration Specification - describes configured system state.

ITRA: IT Risk Assessment - evaluates system-level risks.

ITRRA: IT Requirement Risk Assessment - links requirements to risk.

DIA: Data Integrity Assessment - evaluates data integrity concerns and controls.

TRM: Traceability Matrix - connects requirements, risks, design and verification.

IQ: Installation Qualification - evidence that the system/environment is installed as intended.

VSF: Final Validation Summary Report / Release Recommendation.

ITPSE: IT Periodic System Evaluation - ongoing assessment of system state.

ALCOA+: Data integrity principles covering attributable, legible, contemporaneous, original, accurate, complete, consistent, enduring and available records.

RAG: Retrieval-Augmented Generation: retrieve evidence first, then generate language using that evidence.

Temporal reasoning: Reasoning that uses time applicability, effective dates, supersession and events.

Evidence Graph: Structured relationships connecting evidence objects and lifecycle artifacts.

System State: Canonical structured representation of the system's current or historical assurance condition.

Release Posture: GO / CONDITIONAL / HOLD / DEFERRED decision state; distinct from a numerical assurance indicator.

Prompt injection: Instructions embedded in untrusted content that attempt to manipulate the model. They must be treated as data, not commands.

28. Source Notes

Hackathon / mentor notes: The supplied agenda states the problem as Agentic AI for Always-On, Audit-Ready GxP IT System Management. It highlights lifecycle documentation, historical retrieval, correct document version for a requested timeframe, synthetic data, multiple test scenarios, local prototype execution, and a non-technical presentation approach.

LIMS lifecycle package: The document describes a risk-based lifecycle and inventory model spanning concept, requirements, design, verification, release, operations, periodic evaluation and retirement. It includes requirements, risks, audit trail, access, backup/recovery, configuration, incidents, change, CAPA, training and periodic evaluation records.

GxP Sentinel manual: The reference prototype defines evidence-first review, visible uncertainty, human accountability, trust states, local operation, deterministic findings, evidence traceability, human approval, Assurance Lab tests, and evidence-pack export.

PAS-X dummy package: The supplied files define a fictional NOVOLIFE-MES evidence world with lifecycle, testing, risk, access, audit trail, recovery, change, incident, operations, release, periodic evaluation, training and traceability records.

Current repository assessment: The early repository scaffold contains useful components but also legacy LIMS content, hardcoded RAG answers, generated fallback identifiers, multiple readiness semantics, partially simulated agent traces, and a seeded MES corpus smaller than the complete supplied PAS-X package.

29. One-Page Team Checklist

- ☐ Canonical system = SYS-MES-001 / NOVOLIFE-MES / PAS-X
- ☐ Full PAS-X corpus registered and hashed
- ☐ Evidence schema frozen
- ☐ Traceability relationships evidence-derived
- ☐ Temporal applicability implemented
- ☐ Configurable assurance rules implemented
- ☐ Evidence Agent works
- ☐ Assurance Agent works
- ☐ Risk/Impact Agent works
- ☐ Supervisor uses shared state
- ☐ Dashboard uses live SystemState
- ☐ Copilot cites evidence
- ☐ Release posture is derived, not hardcoded
- ☐ Human approval gateway blocks unsafe actions
- ☐ Conflict/stale/injection safety cases pass
- ☐ Evidence pack reconciles with UI
- ☐ Golden questions pass
- ☐ Local/offline demo passes on clean start
- ☐ Legacy LIMS demo paths removed from the main story
- ☐ Final presentation is simple and outcome-focused

END OF BLUEPRINT